

Question	Answer	Marks
1(a)	rate of change of momentum	B1
1(b)	$\text{kg m s}^{-1} / \text{s} = \text{kg m s}^{-2}$	B1
1(c)	(Base units of) $F = \text{kg m s}^{-2}$ and (Base units of) $m = \text{kg}$ and (Base units of) $A = \text{m}^2$	C1
	(Base units of) RHS = $\text{kg m s}^{-2} \times \text{m}^2 \text{s}^{-1} / (\text{kg} \times \text{m}^2)$ = m s^{-3} (Base units of) LHS = m s^{-1} and (base units of) RHS = m s^{-3} so equation is not homogeneous	A1